Element Learning: Numerical Experiments Report

1D Burgers, 2D Linear Wave, 2D Nonlinear Shallow Water

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Abstract

We report numerical experiments for data-driven surrogate models of three PDEs: (1) 1D viscous Burgers equation (finite-volume reference vs. MLP); (2) 2D linear wave equation (Fourier spectral reference vs. MLP); (3) 2D nonlinear shallow water (Fourier spectral reference vs. UNet). For each problem we generate training data from the reference solver, train a neural network, and compare reference and NN rollouts in time. Training curves and comparison figures are included.

1 Background

1.1 Element learning

The goal is to learn a local map on a small spatial element or patch from data produced by a high-fidelity solver. At inference time, the learned map is applied in rollout form over the full domain, yielding a fast surrogate that can be compared to the reference solver in both accuracy and runtime.

1.2 Three test cases

- **1D Burgers (viscous):** $u_t + uu_x = \nu u_{xx}$ on $[0, L]$ with periodic BC. Reference: finite-volume Godunov flux with implicit diffusion. NN: MLP mapping a 1D patch to the interior window over one NN step.
- **2D linear wave:** $u_{tt} = c^2(u_{xx}+u_{yy})$ with periodic BC. Reference: Fourier pseudo-spectral with RK4 time stepping. NN: MLP mapping patch data from displacement and velocity channels to the interior window.
- **2D nonlinear shallow water:** state (h, q_x, q_y) solved by a spectral reference model. NN: UNet patch-to-patch predictor with residual learning, trained with per-channel absolute TV regularization.

2 Experimental parameters

2.1 1D Burgers

Table 1: Burgers configuration from `config/burgers_1d_config.py`.

Parameter	Value
Domain length L	2.0
Grid cells n_x	2000
CFL	0.5
Patch half-width n_{st}	51
Interior width n_{wd}	100
Frames per NN step n_{jp}	2
Viscosity ν	10^{-3}
IC decay α	2.5
Training samples	5000
Batch size	128
Epochs	2500
Hidden size / layers	128 / 5
Activation	ReLU

2.2 2D linear wave

Table 2: Linear wave configuration from `config/wave_2d_linear_config.py`.

Parameter	Value
Grid $N_X \times N_Y$	256×256
Domain $L_x = L_y$	2π
Δt	2×10^{-3}
Wave speed c	1.0
Steps per saved frame T_{screen}	50
Frames per NN step n_{jp}	2
Patch half-width n_{st}	9
Interior width n_{wd}	32
Training samples	5000
Batch size	128
Epochs	2500
Hidden size / layers	256 / 1
Activation	linear
Compare IC	ring

2.3 2D nonlinear shallow water

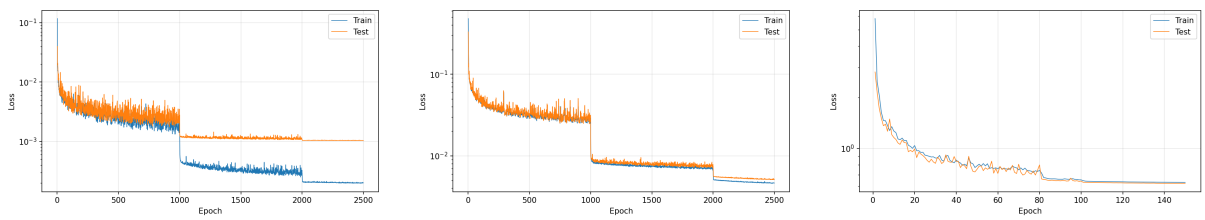
Table 3: Nonlinear shallow-water configuration from `config/wave_2d_nonlinear_config.py`.

Parameter	Value
Grid $n_x \times n_y$	256×256
Domain $L_x = L_y$	2π
g, h_0	9.8, 1.0
Time integrator	IMEX (Strang splitting)
Time step Δt	$0.5 \min(\Delta x, \Delta y)$
Steps per saved frame T_{screen}	10
Patch half-width n_{st}	32
Interior width n_{wd}	32
Viscosity ν_h	0
Nudging coefficient	1.0
Batch size	100
Warmup time	2.0
Compare T_F	8.0
Compare frames	8
Compare IC	random
Training samples	2500
Frames per NN step n_{jp}	2
Epochs	150
Model type	UNet
UNet base channels	32
Pooling	avg
Residual prediction	False
TV smooth weight	[0, 0.1, 0.1]
TV smooth mode	absolute
Data loss channel weights	[1.0, 1.0, 1.0]

3 Results

3.1 Training history

Figure 1 shows the train and test loss versus epoch for the three problems.



(a) Burgers 1D (b) 2D linear wave (c) 2D nonlinear

Figure 1: Training and test loss versus epoch.

3.2 1D Burgers: FV vs. NN

Comparison at 6 output times over $t \in [0, 2]$ with `compare_t_end=2.0`. Runtime for this run: FV total 2.778 s, NN total 0.5852 s, speedup 4.7×.

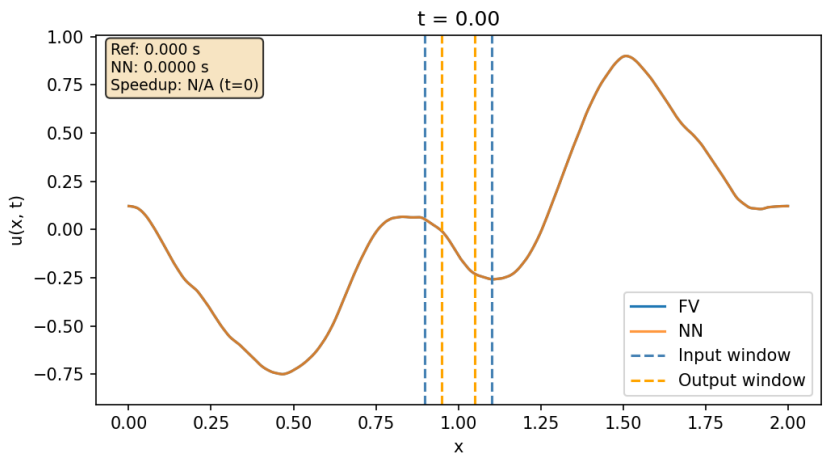


Figure 2: Burgers: initial frame.

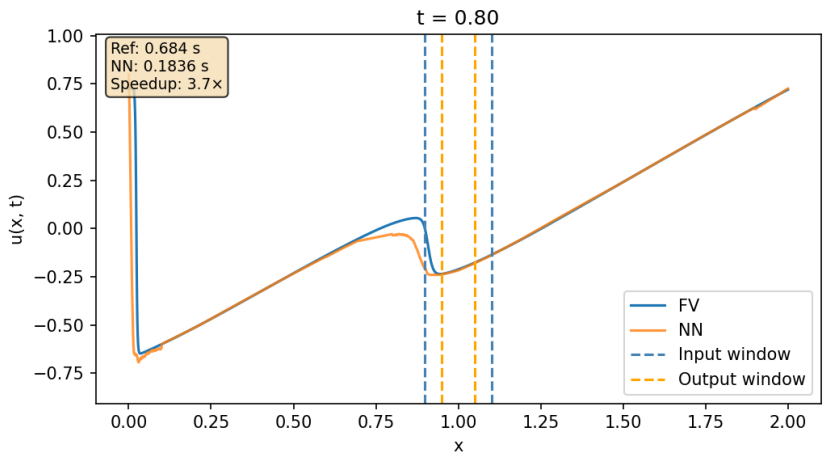


Figure 3: Burgers: mid-run frame.

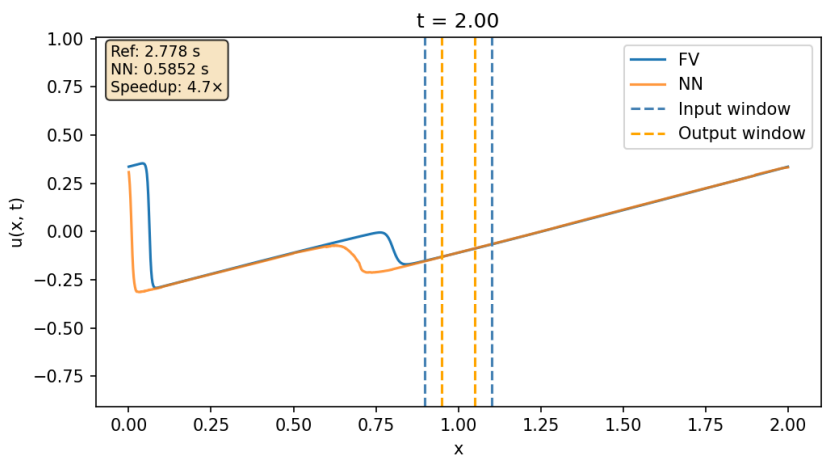


Figure 4: Burgers: final frame.

3.3 2D linear wave: Spectral vs. NN

Comparison at 6 aligned frames with `compare_TF=2.0` and `compare_ic=ring`. Runtime for this run: spectral 10.323 s, NN 0.4576 s, speedup 22.6×.

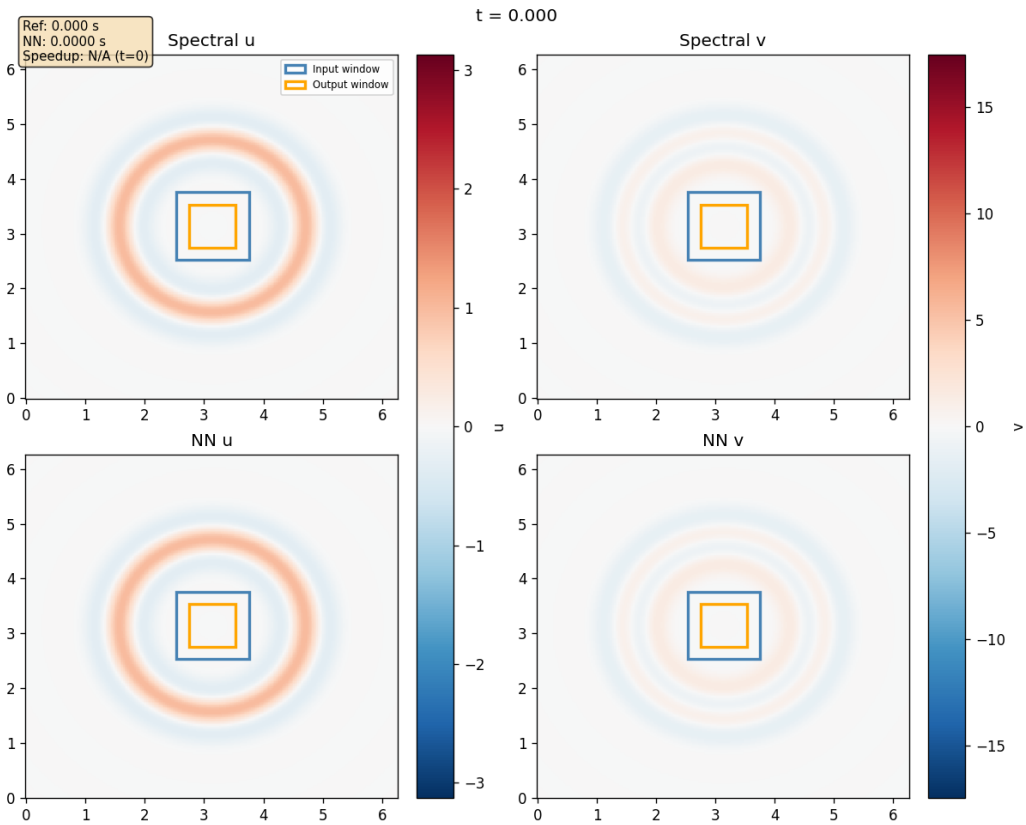


Figure 5: 2D linear wave: initial frame.

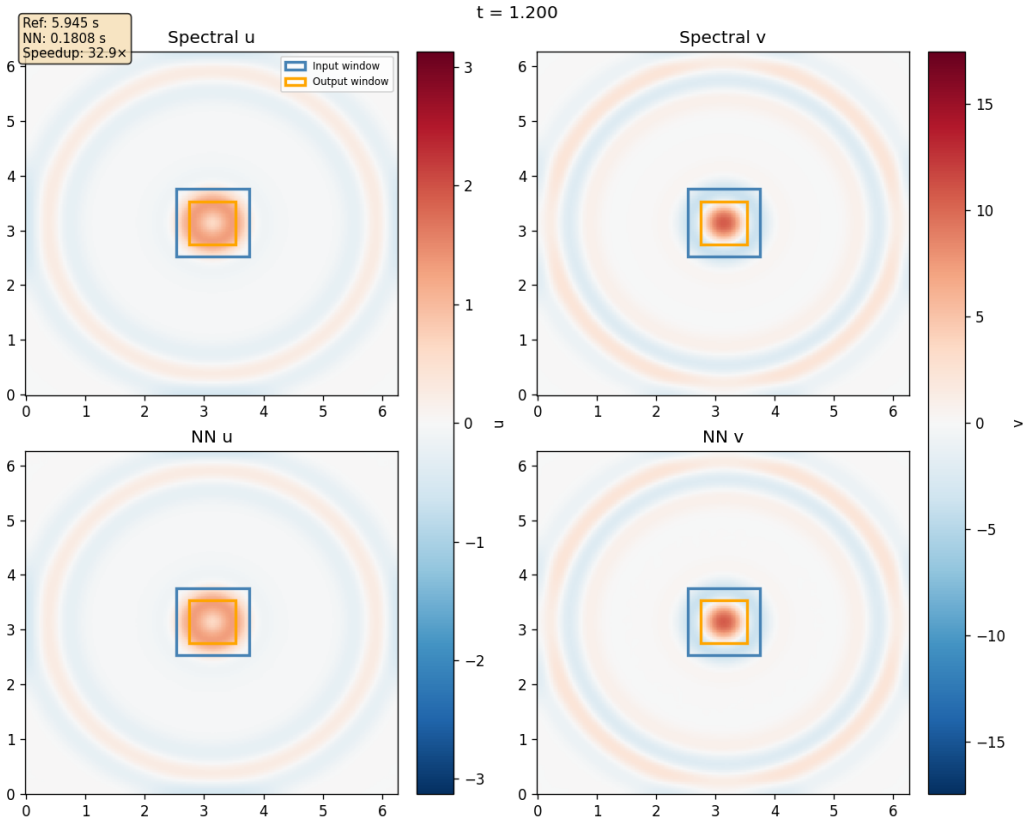


Figure 6: 2D linear wave: mid-run frame.

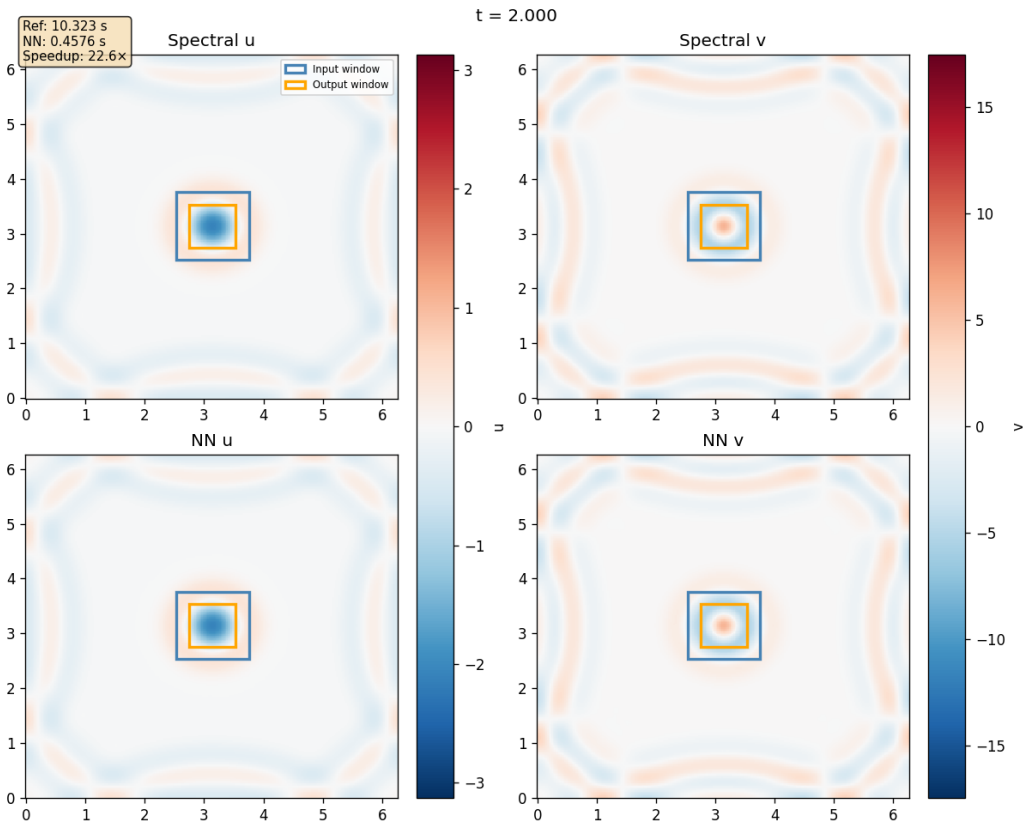


Figure 7: 2D linear wave: final frame.

3.4 2D nonlinear shallow water: Spectral vs. NN

Comparison at 8 aligned frames for $(h-h_0, q_x, q_y)$ with `compare_TF=8.0` and `compare_ic=random`. Both rollouts start after warmup time $T = 2.0$. Final rollout metrics for this run: mean L1 error 0.047030, max-frame L1 error 0.075241, spectral runtime 161.058 s, NN runtime 10.251 s, speedup 15.7×.

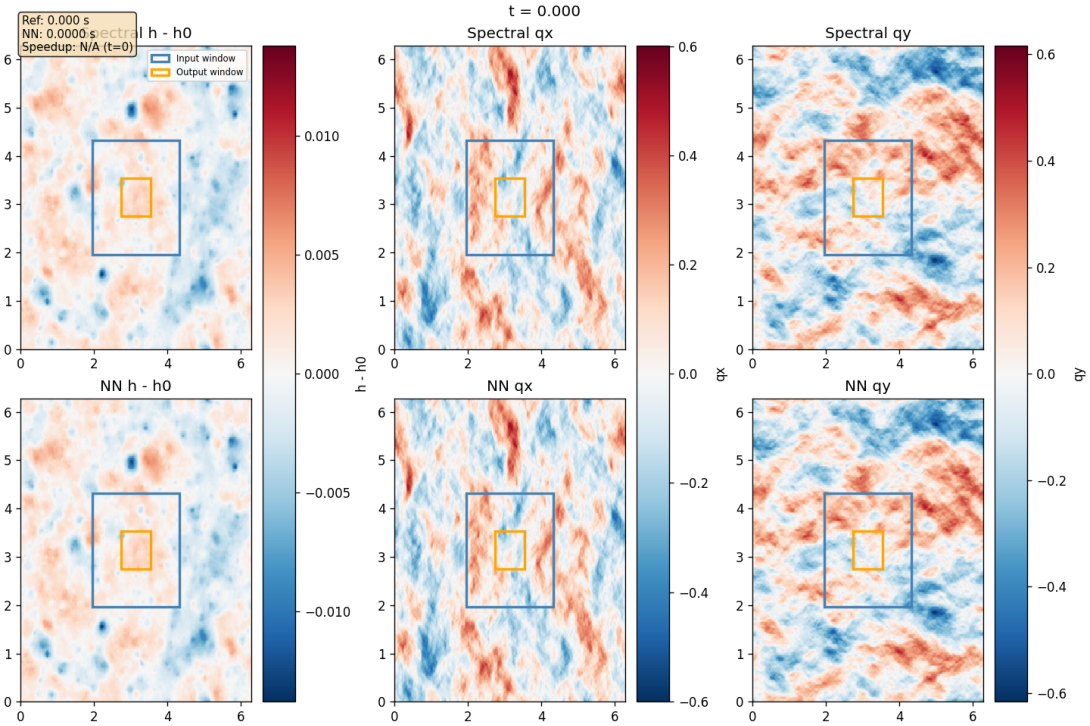


Figure 8: 2D nonlinear shallow water: initial frame.

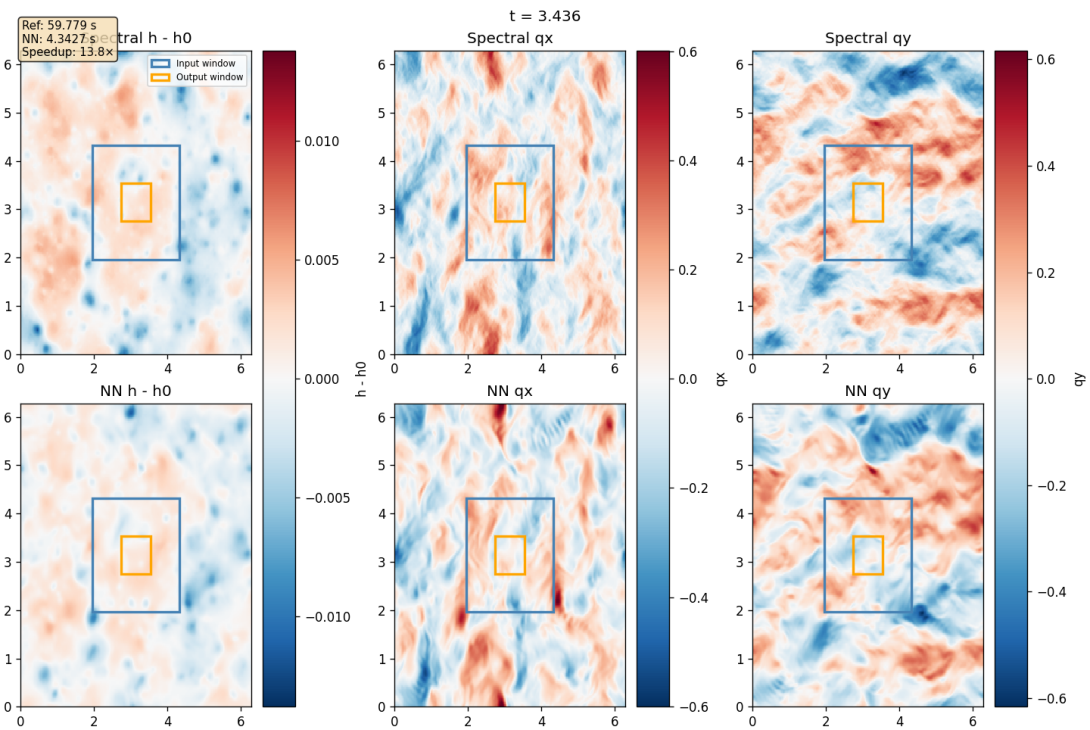


Figure 9: 2D nonlinear shallow water: mid-run frame.

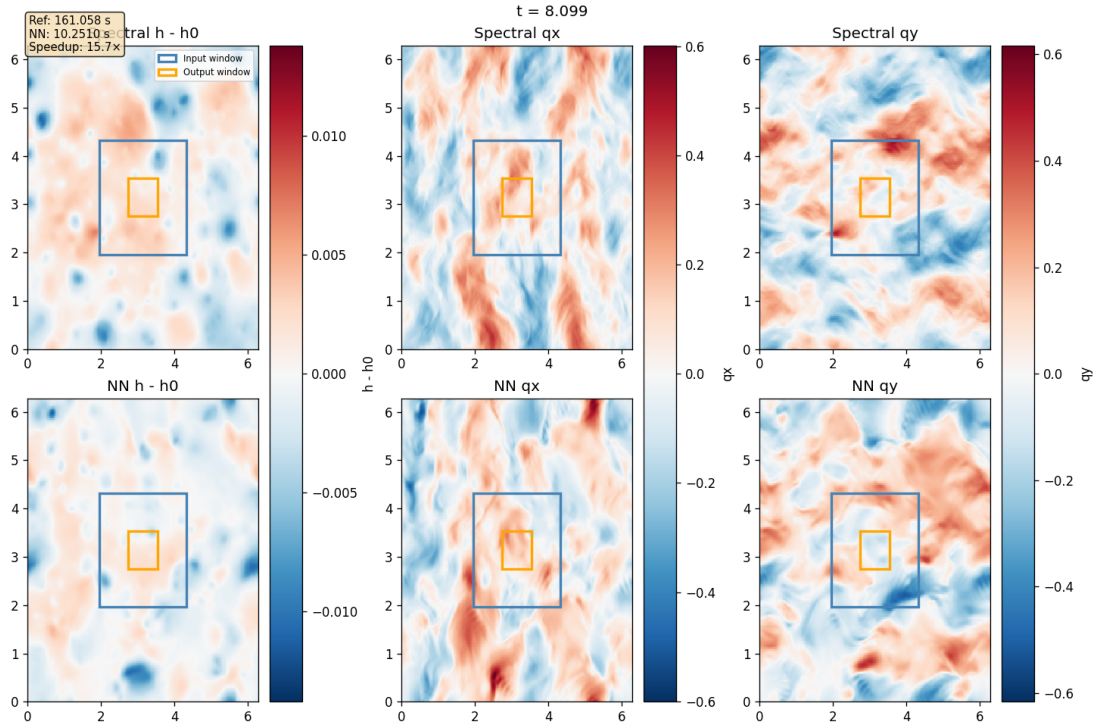


Figure 10: 2D nonlinear shallow water: final frame.

4 Conclusion

The three pipelines run successfully with a unified workflow: `gen_data.py`, `python -m ml.train`, and `compare.py` are all selected through the shared `-problem` interface. The rebuilt figures show that the learned surrogates track the reference solutions while delivering clear runtime savings in all three cases.